

MATH 3MB3 PROJECT REPORT

EBOLA TRANSMISSION REPORT

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1 Introduction

The repeated outbreaks of global infectious disease pandemics over the past few years have intensified the necessity for data and model studies. This study chooses the Ebola virus as objects of study due to its typically high mortality rate, transmission rate, and present-day possibility of control through public health interventions. With empirical datasets of Ebola virus disease outbreaks, we build a nonlinear dynamical model for the prediction of long-term behavior of epidemic spreading.

Ebola virus disease (EVD), clinically known as Ebola hemorrhagic fever, is an acute infectious disease characterized by intense contagion and death[1]. Originally recognized during concurrent 1976 outbreaks in Nzara (now South Sudan) and Yambuku (Democratic Republic of the Congo), recurring epidemics—primarily in sub-Saharan Africa—have regularly threatened global health security [2]. A particular upsurge between March and August of 2014 saw 1,603 suspected and confirmed cases reported from Guinea, Sierra Leone, Liberia, and Nigeria, of which 887 died—a fatality rate of around 55% [3].

It is transmitted via contact with infected body fluids from people with symptoms. At the asymptomatic incubation stage, limited viral presence makes the transmission of the disease close to zero [2]. When the person gets sick with diarrhoea and vomiting among other symptoms, body fluids of all kinds are infectious, especially the blood, faeces and vomit that can harbor elevated concentrations of the virus [2].

Additionally, cultural practices also have a critical impact on transmission dynamics. Traditional funeral rites of washing corpses, tactile contact (kissing and touching of the deceased) and ritual hugs remain prevalent in Guinea and other West African states amid epidemiological risks [4]. The rites present immediate contact with virulent body fluids, making postmortem transmission a pathway with cultural mediation unique in EVD epidemiology[4].

Methodologically, we built upon basic Susceptible-Infected-Removed (SIR) models during lecture. In this paper, the modification of the basic model is adding a funeral transmission compartment (F), creating a SIFR base framework. Addition of Exposed (E) and Hospitalized (H) compartments later provides the advanced SHERIF model. Comparison of SIFR and SHERIF

model predictions will measure their relative ability to predict the course of an epidemic, with specific focus on projected model fit in cultural environments.

2 Model

To simulate the mechanism of the spread of Ebola virus among humans in West Africa, we set up two dynamical, non-linear, multivariate, deterministic, and discrete models respectively, which are SIFR model and SHERIF model. These models include not only basic social mechanisms of transmission, such as spreading through direct contact with infected people or by touching exposed dead bodies during a funeral, but also biological and cultural factors such as funerary behavior, public health interventions, mother-to-child transmission, and incubation period.

2.1 Base Model (SIFR Model)

We aim to model the recent African Ebola outbreak in a population using the SIFR model, which accounts for transmission caused by contact during funerals.

2.1.1 SIFR Model Assumption

We made appropriate simplifications of reality when building the models, the following will introduce the basic assumptions contained in the base model.

1. In this model, we classify people into 4 distinct groups: S, R, I, and F, and each individual only belongs to one group in each period (week).
2. This model considers the transmission of only the Ebola virus, including spread during funerals (F).
3. Closed population, with no external population such as immigration.
4. S and I have the same birth rate (b), and Ebola is transmitted from mother to child.
5. I can only transition to F through the death of Ebola, then from F to

R. We assume there is no medical cure for Ebola in this model.

6. Only I and F are infectious. Other groups contacting I and F are infected by a certain probability.
7. Properly handled bodies are non-infectious; funeral-exposed bodies are infectious.
8. This model only considers funerals for individuals who died of Ebola. We assume all individuals who die of Ebola hold a funeral, and only Ebola-related funerals are included in this model. Also, in our model, each individual is assumed to have at most one funeral.
9. Recovered individuals do not experience reinfection (except for sexual transmission, which is not considered). R will never be reinfected in this model.

2.1.2 State Variable Notation

Table 1 summarizes the state variables involved in the model that are used to characterize the dynamics of the process at different moments in time.

Symbol	Definition	Unit
S	Susceptible: individuals who are not infected but at a risk of infection ($0 \leq S$)	Numbers of individuals in million
I	Infectious: infected also infectious individuals ($0 \leq I$)	Numbers of individuals in million
F	Funeral: individuals who have died from infection and are currently holding funerals ($0 \leq F$)	Numbers of individuals in million
R	Removed: individuals who died and are no longer infectious or susceptible ($0 \leq R$)	Numbers of individuals in million

Table 1: SIFR Model State Variable Notation

2.1.3 Parameter Notation

Table 2 summarizes the parameters involved in the model that are used to characterize the dynamics of the process at different moments in time.

Symbol	Definition	Unit
b	Birth rate ($0 \leq b$)	per individual per week
d	Natural death rate ($0 \leq d \leq 1$)	per individual per week
β	Transmission probability per contact between S and I ($0 \leq \beta \leq 1$)	per contact per week
β_F	Transmission probability per contact between S and F ($0 \leq \beta_F \leq 1$)	per contact per week
γ	Probability to die of Ebola ($0 \leq \gamma \leq 1$)	per week
σ	Transition probability from I to F ($0 \leq \sigma \leq 1$)	per week
N	Initial population size	Numbers of individuals in million
S_0	Initial number of S	Numbers of individuals in million
I_0	Initial number of I	Numbers of individuals in million
F_0	Initial number of F	Numbers of individuals in million
R_0	Initial number of R	Numbers of individuals in million

Table 2: SIFR Model Parameter Notation

2.1.4 SIFR Model Equation

This SIFR model is a dynamical, non-linear, multivariate, deterministic, and discrete model, which is represented as Equation 1

$$\begin{cases} S_{t+1} = S_t - \beta S_t I_t - \beta_F S_t F_t + b S_t - d S_t \\ I_{t+1} = I_t + \beta S_t I_t + \beta_F S_t F_t - \gamma I_t + b I_t \\ F_{t+1} = F_t + \gamma I_t - \sigma F_t \\ R_{t+1} = R_t + \sigma F_t + d S_t \end{cases} \quad (1)$$

2.1.5 SIFR Model Flow Diagram

For clear analysis, we simulated the flow diagram between populations and Ebola transmission based on assumptions and model equations. The flow of this model is shown in Figure 1.

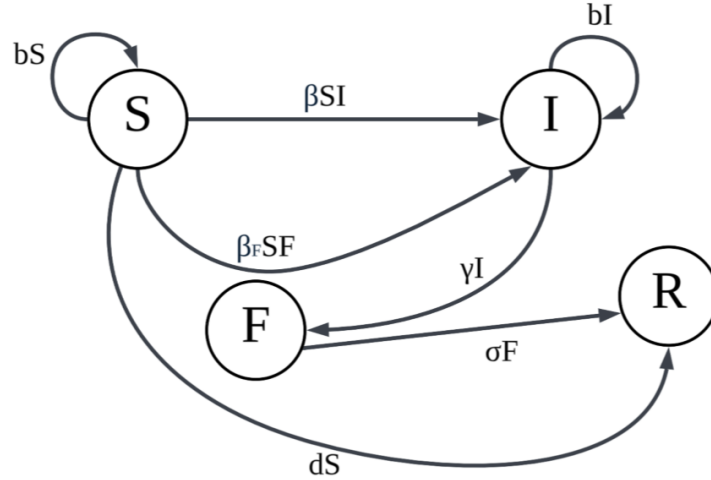


Figure 1: SIFR Model Flow Diagram

We outline flows of transitions between each state and explain how the parameters determine the rate and probability of these transitions.

- $S \rightarrow I$ (Susceptible to Infectious):

1. After contacting infectious individuals, susceptible individuals get infected at a probability of β .
 2. After attending funerals and contacting dead bodies, susceptible individuals get infected at a probability of β_F .
- $S \rightarrow S$ (Susceptible to Susceptible): Newborns inherit their infection status from their mothers. They are born to a susceptible individual, so they remain susceptible.
 - $S \rightarrow R$ (Susceptible to Removed): d of susceptible individuals naturally died without getting an infection, then they transit to R.
 - $I \rightarrow F$ (Infectious to Funeral): Infectious individuals die of Ebola and hold funerals, with a weekly probability γ .
 - $I \rightarrow I$ (Infectious to Infectious): If babies are born to an infected individual, they are already infected.
 - $F \rightarrow R$ (Funeral to Removed): Every week, σF funerals end and their bodies will be properly handled to remove the risk of infection.

2.2 Analysis of Base Model

we can simulate the transmission of Ebola according to Equation 1. By definition, the equilibrium point is the point where the state variable stops changing from one step to the next, denoted by (S^*, I^*, F^*, R^*) . After some algebra, the equilibrium points for this model are $(S^*, I^*, F^*, R^*) = (0, 0, 0, R^*)$ and $(S^*, I^*, F^*, R^*) = (\frac{-\sigma\gamma(b-d)}{d(\beta\sigma+\beta_F\gamma)}, \frac{d-b}{-\beta-\frac{\beta_F\gamma}{\sigma}}, \frac{\gamma(d-b)}{\beta\sigma-\beta_F\gamma}, R^*)$, where R^* is a nonnegative value that represents individuals who have recovered or died. Since the feasible range for state variables and parameters are nonnegative, the only reasonable equilibrium point for this model is $(S^*, I^*, F^*, R^*) = (0, 0, 0, R^*)$. This equilibrium point suggests the disease-free state, where everyone in the community has either died from Ebola. So, no further transmission can occur in this system.

$$J = \begin{bmatrix} 1 - \beta I_t - \beta_F F_t + b - d & -\beta S_t & -\beta_F S_t & 0 \\ \beta I_t + \beta_F F_t & 1 + \beta S_t - \gamma + b & \beta_F S_t & 0 \\ 0 & \gamma & 1 - \sigma & 0 \\ d & 0 & \sigma & 1 \end{bmatrix} \quad (2)$$

$$J|_{(0,0,0,N)} = \begin{bmatrix} 1 + b - d & 0 & 0 & 0 \\ 0 & 1 - \gamma + b & 0 & 0 \\ 0 & \gamma & 1 - \sigma & 0 \\ d & 0 & \sigma & 1 \end{bmatrix} \quad (3)$$

After some calculations, the Jacobian matrix for this model is shown in Equation 2. So, as shown in Equation 3, $J|_{(0,0,0,R^*)}$ is a lower triangular matrix, which indicates the eigenvalues are the same as the diagonal entries of the matrix. So, the eigenvalues for this model are $\lambda_1 = 1 + b - d$, $\lambda_2 = 1 - \gamma + b$, $\lambda_3 = 1 - \sigma$, $\lambda_4 = 1$. After some algebra, λ_1, λ_2 and λ_3 is less than 1 if $0 < b < d$ and $b < \gamma < 1$. The equilibrium point is stable if the magnitude of the eigenvalue is less than 1, but λ_4 always equals to 1. So in this situation, we need to take a closer look for stability of this equilibrium point. Otherwise, this equilibrium point is unstable for this model.

Table 3: Base Model Parameters under Stable Conditions

Symbol	Value
N	10
b	0.0005
d	0.0006
β	0.1
β_F	0.3
γ	0.5
σ	0.5
S_0	9.99
I_0	0.01
F_0	0
R_0	0

To better understand this base model, we visualize the model simulation using R as Figure 2, based on the parameters in Table 4. Because of the lack

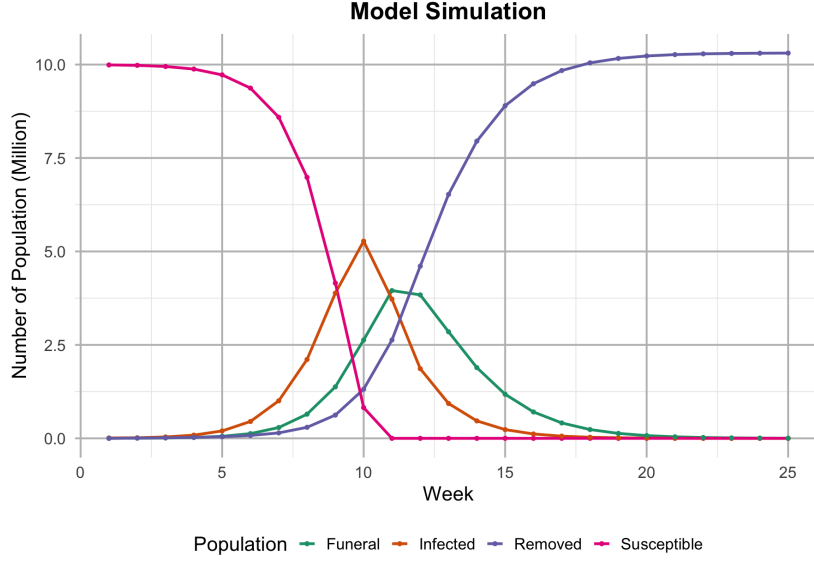


Figure 2: Base Model Simulation

of data in real time, the parameters were chosen with reference to the characteristics of Ebola transmission, the high fatality rate, and funeral customs in Africa. We notice that the parameters fits the stable condition for partial eigenvalues, which means the magnitude of λ_1 , λ_2 and λ_3 are less than 1 under these parameters.

This figure shows the simulation results of the basic Ebola transmission model over a 25-week period. At first, almost the entire population is susceptible, and only a very small number of people are infected. With the spread of the virus, the number of susceptible people drops quickly, and the size of Funeral(F) population gradually increase. The pandemic reaches its peak at week 10, then the number of infected people starts to decrease. By week 21, almost all population enter the Removed(R) state, with each of the other state variables approaching to zero. The simulation result indicates that under the current model structure and parameter setting, the system experiences a complete transmission of Ebola, and finally the whole population is infected and died. Therefore, the system approaches to a state where there are not infections. This is consistent with the equilibrium point $(S^*, I^*, F^*, R^*) = (0, 0, 0, R^*)$. Therefore, this equilibrium point is stable under these parameters.

Table 4: Base Model Parameters under Unstable conditons

Symbol	Value
N	10
b	0.000373[5]
d	0.000231[6]
β	0.1
β_F	0.3
γ	0.3
σ	0.5
S_0	9.99
I_0	0.01
F_0	0
R_0	0

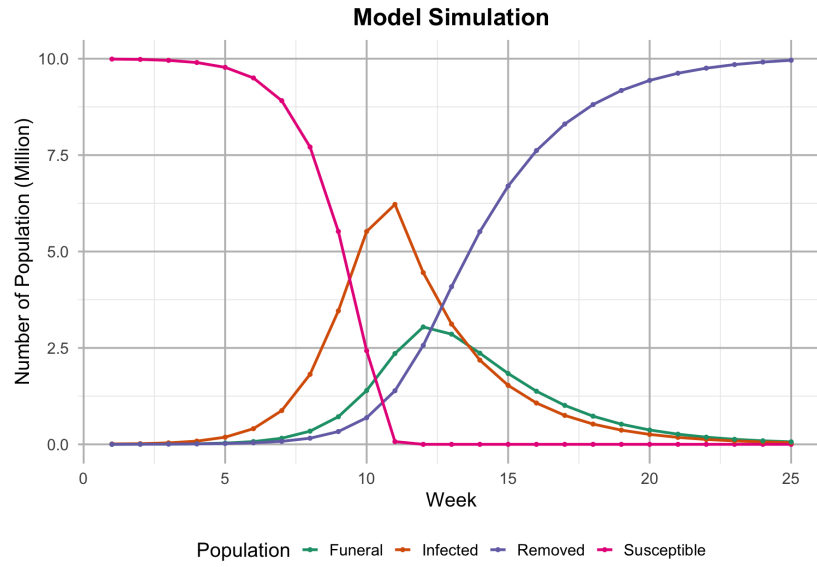


Figure 3: Base Model Simulation

Then, in order to better explore the long-time behavior of this base model, we choose parameter value outside stable conditions, which as shown in Table 3. The simulation result is shown as Figure 3. We find that this results is very similar with the previous one. This indicates the equilibrium point $(S^*, I^*, F^*, R^*) = (0, 0, 0, R^*)$ is also stable under these parameters. Therefore, we conclude that the equilibrium point $(S^*, I^*, F^*, R^*) = (0, 0, 0, R^*)$ is always stable.

2.3 Model Extension (SHERIF Model)

In order to control the spread of Ebola and improve the medical level, we add state variables of E (Exposed) and H (Hospitalized) to make the model closer to real life. Now the general structure of this model becomes $S \rightarrow E \rightarrow I \rightarrow H \rightarrow F \rightarrow R$. Similarly, we improve the Base model from the following six aspects.

2.3.1 SHERIF Model Assumption

We made appropriate simplifications of reality when building the models, the following will introduce the basic assumptions contained in the extended model.

1. In this model, we classify people into 6 distinct groups: S, H, E, R, I, and F, and each individual only belongs to one group in each period (week).
2. This model considers the transmission of only the Ebola virus, including hospital treatment (H) and spread during funerals (F).
3. Closed population, with no external population such as immigration.
4. S and I have the same birth rate (b), and Ebola is transmitted from mother to child.
5. Due to the short incubation period of Ebola, births in E are not considered in this model.
6. I transition to R through self-recovery, hospital recovery, or death.

7. Transmission follows $S \rightarrow E \rightarrow I$. All infections pass through E, except for mother-to-child transmission.
8. Only I and F are infectious. Other groups contacting I and F are infected by a certain probability.
9. Properly handled bodies are non-infectious; funeral-exposed bodies are infectious.
10. This model only considers funerals for individuals who died of Ebola. We assume all individuals who die of Ebola hold a funeral, only Ebola-related funerals are included in this model. Also, in our model, each individual is assumed to have at most one funeral.
11. Recovered individuals do not experience reinfection (except for sexual transmission, which is not considered). R will never be reinfected in this model.
12. To focus on the spread of Ebola rather than population changes, births from recovered individuals in R are not included in this model.

2.3.2 State Variable Notation

Table 5 summarizes the state variables involved in the model that are used to characterize the dynamics of the process at different moments in time.

2.3.3 Parameter Notation

Table 6 summarizes the parameters involved in the model that are used to characterize the dynamics of the process at different moments in time.

2.3.4 SHERIF Model Equation

This SHERIF model is a dynamical, non-linear, multivariate, deterministic, and discrete model, which is represented as Equation 4

Symbol	Definition	Unit
S	Susceptible: individuals who are not infected but at a risk of infection ($0 \leq S$)	Numbers of individuals in million
H	Hospitalized: infected individuals who are hospitalized (for treatment) ($0 \leq H$)	Numbers of individuals in million
E	Exposed: infected individuals but not yet infectious.(individuals exposed to Ebola and infected, but they are still in the incubation period and are not yet infectious) ($0 \leq E$)	Numbers of individuals in million
R	Removed: individuals who have recovered or died and are no longer infectious or susceptible ($0 \leq R$)	Numbers of individuals in million
I	Infectious: infected also infectious individuals ($0 \leq I$)	Numbers of individuals in million
F	Funeral: individuals who have died from infection and are currently holding funerals ($0 \leq F$)	Numbers of individuals in million

Table 5: SHERIF Model State Variable Notation

Symbol	Definition	Unit
b	Birth rate ($0 \leq b$)	per individual per week
d	Natural death rate ($0 \leq d \leq 1$)	per individual per week
β	Transmission probability per contact between S and I ($0 \leq \beta \leq 1$)	per contact per week
β_F	Transmission probability per contact between S and F ($0 \leq \beta_F \leq 1$)	per contact per week
γ_1	Probability to die of Ebola without hospitalized ($0 \leq \gamma_1 \leq 1$)	per week
γ_2	Probability to die of Ebola with hospitalized ($0 \leq \gamma_2 \leq 1$)	per week
σ_1	Transition probability from E to I ($0 \leq \sigma_1 \leq 1$)	per week
σ_2	Transition probability from F to R ($0 \leq \sigma_2 \leq 1$)	per week
p_1	Recovery probability from I to R ($0 \leq p_1 \leq 1$)	per week
p_2	Recovery probability from I to H ($0 \leq p_2 \leq 1$)	per week
h	Hospitalized probability of I ($0 \leq h \leq 1$)	per week
N	Initial population size	Numbers of individuals in million
S_0	Initial number of S	Numbers of individuals in million
H_0	Initial number of H	Numbers of individuals in million
E_0	Initial number of E	Numbers of individuals in million
R_0	Initial number of R	Numbers of individuals in million
I_0	Initial number of I	Numbers of individuals in million
F_0	Initial number of F	Numbers of individuals in million

Table 6: SHERIF Model Parameter Notation

$$\begin{cases} S_{t+1} = S_t - \beta S_t I_t - \beta_F S_t F_t + b S_t - d S_t \\ H_{t+1} = H_t + h I_t - \gamma_2 H_t - p_2 H_t \\ E_{t+1} = E_t + \beta S_t I_t + \beta_F S_t F_t - \sigma_1 E_t \\ R_{t+1} = R_t + p_1 I_t + p_2 H_t + \sigma_2 F_t + d S_t \\ I_{t+1} = I_t + \sigma_1 E_t - \gamma_1 I_t - h I_t - p_1 I_t + b I_t \\ F_{t+1} = F_t + \gamma_1 I_t + \gamma_2 H_t - \sigma_2 F_t \end{cases} \quad (4)$$

2.3.5 SHERIF Model Flow Diagram

For clear analysis, we simulated the flow diagram between populations and Ebola transmission based on assumptions and model equations. The flow of this model is shown in Figure 4.

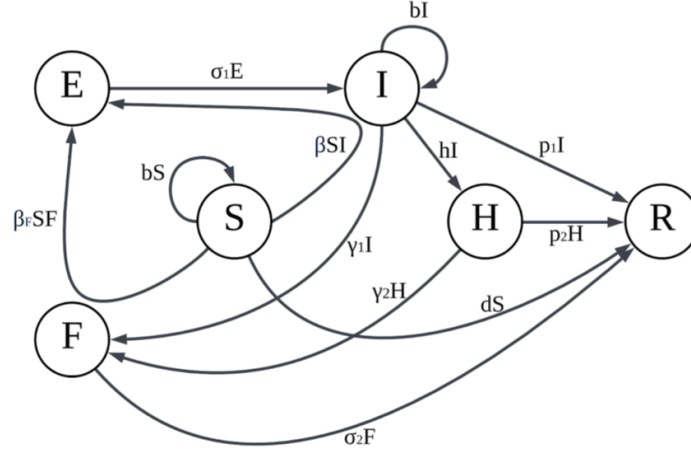


Figure 4: SHERIF Model Flow Diagram

We outline flows of transitions between each state and explain how the parameters determine the rate and probability of these transitions.

- $S \rightarrow E$ (Susceptible to Exposed):
 1. After contacting infectious individuals, susceptible individuals get infected at a probability of β .
 2. After attending funerals and contacting dead bodies, susceptible

individuals get infected at a probability of β_F .

- $S \rightarrow S$ (Susceptible to Susceptible): Newborns inherit their infection status from their mothers. They are born to a susceptible individual, so they remain susceptible.
- $S \rightarrow R$ (Susceptible to Removed): d of susceptible individuals naturally died without getting an infection, then they transit to R.
- $E \rightarrow I$ (Exposed to Infectious): After the incubation period, exposed individuals progress to infectious individuals with a weekly transition probability σ_1 . At this stage, they become symptomatic and infectious.
- $I \rightarrow F$ (Infectious to Funeral): Infectious individuals die of Ebola and hold funerals, with a weekly probability γ_1 .
- $I \rightarrow H$ (Infectious to Hospitalized): Every week, hI infectious individuals seek hospital treatment.
- $I \rightarrow R$ (Infectious to Removed): Every week, p_1I infectious individuals recover without hospitalization.
- $I \rightarrow I$ (Infectious to Infectious): If babies are born to an infected individual, they are already infected.
- $H \rightarrow F$ (Hospitalized to Funeral): Every hospitalized patient has a weekly probability γ_2 of not recovering and dying, then holding funerals.
- $H \rightarrow R$ (Hospitalized to Removed): There are p_2H individuals recovering from the hospital and becoming removed individuals every week.
- $F \rightarrow R$ (Funeral to Removed): Every week, σ_2F funerals end and their bodies will be properly handled to remove the risk of infection

In the next section, we will apply the model to real-life situations with data. We will also predict and analyze the long-term behavior of Ebola transmission. In this way, we can not only verify the validity of the model but also explore the potential impact of health interventions on the Ebola virus.

3 Results

Based on Equation 4, we can simulate the transmission of the Ebola virus within a specific closed community under the assumption. We denote the total population within that community at the beginning by N .

First, we try to find the equilibrium point where the state variable stops changing from one time step to the next. We denote the equilibrium point by $(S^*, E^*, I^*, H^*, F^*, R^*)$. After some algebra and considering the feasible range of state variables, we find that the only reasonable equilibrium point in this model is $(S^*, E^*, I^*, H^*, F^*, R^*) = (0, 0, 0, 0, 0, R^*)$, where R^* is a nonnegative value that represents individuals who have recovered or died. This equilibrium point suggests the disease-free state, where everyone in the community has either died from Ebola or recovered and gained permanent immunity. As a result, no further transmission can occur in this system.

$$J = \begin{bmatrix} -\beta_F F_t - \beta I_t + b - d + 1 & 0 & -\beta S_t & 0 & -\beta_F S_t & 0 \\ \beta_F F_t + \beta I_t & 1 - \sigma_1 & \beta S_t & 0 & \beta_F S_t & 0 \\ 0 & \sigma_1 & b - h - p_1 + \sigma_1 - \gamma_1 + 1 & 0 & 0 & 0 \\ 0 & 0 & h & 1 - \gamma_2 - p_2 & 0 & 0 \\ 0 & 0 & \gamma_1 & \gamma_2 & 1 - \sigma_2 & 0 \\ d & 0 & p_1 & p_2 & \sigma_2 & 1 \end{bmatrix} \quad (5)$$

Table 7: Model Fixed Parameters

Symbol	Value
b	0.000373 [5]
d	0.000231 [6]
β	0.1
β_F	0.3
γ_1	0.7
σ_1	0.7
σ_2	0.5
p_1	0.01

After that, we check the stability of this equilibrium point to determine whether our model is actually close to this point as time goes on. After some calculations, the Jacobian matrix for our model is shown in Equation 5. So, as shown in Equation 6, $J|_{(0,0,0,0,0,R^*)}$ is a lower triangular matrix, which

indicates that the diagonal entries are the same as the eigenvalues of the matrix. To simplify the analysis, we fix some parameters and focus on the parameter that is modifiable by public health policy, such as h , γ_2 , p_2 . The values of the parameters in the model are shown in Table 7 and remained constant during the whole simulation process. After fixing the parameters, we get the Jacobian matrix $J|_{(0,0,0,0,R^*)}$ as shown in Equation 7.

$$J|_{(0,0,0,0,R^*)} = \begin{bmatrix} b-d+1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1-\sigma_1 & 0 & 0 & 0 & 0 \\ 0 & \sigma_1 & b-h-p_1-\sigma_1+1 & 0 & 0 & 0 \\ 0 & 0 & h & 1-\gamma_2-p_2 & 0 & 0 \\ 0 & 0 & \gamma_1 & \gamma_2 & 1-\sigma_2 & 0 \\ d & 0 & p_1 & p_2 & \sigma_2 & 1 \end{bmatrix} \quad (6)$$

$$J|_{(0,0,0,0,N)} = \begin{bmatrix} 1.000142 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 & 0 & 0 \\ 0 & 0.7 & 0.990373-h & 0 & 0 & 0 \\ 0 & 0 & h & 1-\gamma_2-p_2 & 0 & 0 \\ 0 & 0 & 0.7 & 0.4 & 0.5 & 0 \\ 0.000231 & 0 & 0.01 & 0.5 & 0.5 & 1 \end{bmatrix} \quad (7)$$

By the definition of dominant eigenvalue, the magnitude of the dominant eigenvalue of $J|_{(0,0,0,0,R^*)}$ is greater than 1. Therefore, by mathematical analysis, the equilibrium point $(S^*, E^*, I^*, H^*, F^*, R^*) = (0, 0, 0, 0, 0, R^*)$ is not asymptotically stable. This suggests that mathematically, if you make a very small perturbation near this equilibrium point, the state of the system will move away from the equilibrium point.

However, according to the model assumption, the population is closed, which means that there is no inflow of foreign individuals or infected individuals, and all infected individuals either recover from the infection and obtain permanent immunity or they die. So, if the system reaches this equilibrium point $(S^*, E^*, I^*, H^*, F^*, R^*) = (0, 0, 0, 0, 0, R^*)$, it means that all individuals either recovered or died. Also, there is no source of infection, so the infection will not reoccur. From this perspective, although mathematically that equilibrium point is labeled as unstable, in reality, under the assumptions of the model, once the system reaches this equilibrium state, it cannot move away and no new infections will occur. Therefore, the equilibrium point is considered stable for this model under these assumptions.

Table 8: Model Parameters

Symbol	Value
N	10
h	0.7 [7]
γ_2	0.4
p_2	0.5
S_0	9.99
E_0	0
I_0	0.01
H_0	0
F_0	0
R_0	0

To better understand this extended model, we use the parameters in Table 7 and Table 8 to visualize the model simulation using R as Figure 5. Because of the lack of monitoring data in real time, the parameters were chosen with reference to the characteristics of Ebola transmission, the high fatality rate, and funeral customs in Africa. This figure shows the simulation results of the Ebola transmission model over a 30-week period. At first, almost the entire population is susceptible, and only a very small number of people are infected. With the spread of the virus, the number of susceptible people decreases quickly, and the number of Exposed(E), Infected(I), Hospitalized(H), and Funeral(F) people gradually increase. The pandemic reaches its peak around week 16, then the number of infected people starts to decrease and the number of exposed people gradually reduces. By week 25, almost all populations enter the Removed(R) state, with each of the other state variables approaching to zero. The simulation result indicates that under the current model structure and parameter settings, the system experiences a complete transmission of Ebola, and finally the whole population is infected and immunized. Therefore, the system approach to a state in which no new infections occur, which is consistent with the equilibrium point $(S^*, E^*, I^*, H^*, F^*, R^*) = (0, 0, 0, 0, 0, R^*)$.

Compared with Figure 2 and Figure 3, the increase in the infected population is restrained. In the base model, the size of the infected population reaches the peak quickly. On the other hand, the number of infected people in the extended model increase gradually and its peak is much smaller than the

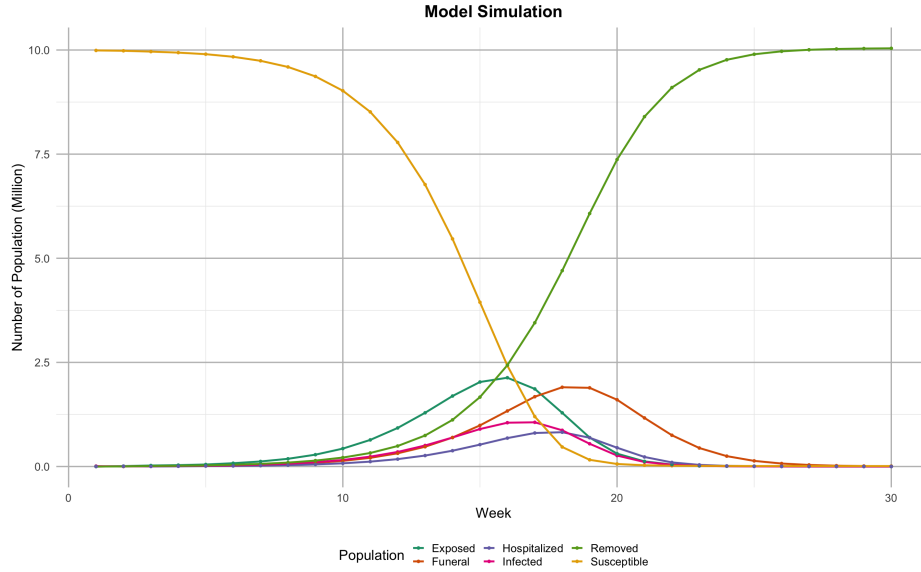


Figure 5: Extended Model Simulation

peak in the base model. Therefore, the extended model is more consistent with real-world situations.

4 Discussion

Based on the model presented in the report and the analysis of the model, it is quite helpful in eliciting the results and the subsequent discussion. Meanwhile, the purpose of this study is to confirm the prediction of the outcome of the long-term Ebola virus epidemic. A brief summary of the results obtained is as follows: According to the model and the analysis in the study, under the constraints of the model, there is a tendency for the mortality rate of Ebola infected people to exceed the infection rate, which means that the infected people will only have two scenarios, surviving or dying. Combined with some of the realities on the ground in Africa, it can be concluded that the particular funeral culture inherited in Africa is a huge contributing factor to the spread of the Ebola virus. With this conclusion, without discussing the human significance of this culture, the medical significance in the context of a pandemic disease is simply discussed. With the unknown factor of death, close contact with corpses and unprotected low health care awareness

is undoubtedly very dangerous. It is hoped that this study will show people the risk of infectious diseases and the importance of good hygiene practices is also an important significance.

When this study was first prepared for modeling, it did not take into account the local African culture and funeral practices. After these were examined this variable was added and re-modeled, but again this modeling was conditional. In the modeling conditions of this study, people who die from Ebola infection must have a funeral, and new people will be infected at the funeral. Therefore, this model is not applicable to other scenarios. For example, in areas where there is no funerary culture of close contact with the dead, in areas where medical protection is more advanced, or where Ebola-infected corpses are subjected to special treatment such as cremation. None of these scenarios and conditions are applicable to the mods of this study. Therefore, whether or not a funeral will be held and whether or not there will be close contact with the corpse at the funeral becomes a key factor in this model, and variations in this factor will affect the prediction of the results.

In the process of modeling and analysis, some extended questions were raised: whether people who survived the Ebola virus infection are still infectious, and whether these survivors are still fertile. Some of these queries can be answered after some investigations. Some of the survivors could still be detected with Ebola 115 days after recovery and Ebola did not destroy their reproductive function[8] [9]. This also raises new questions about whether the survivors' children are also carrying Ebola. But these are additional questions based on the extension of the study. This study is just a study based on the modeling of Ebola virus transmission in the particular environment of Africa. However, if other conditions or factors change, the analysis in this study can still be used as a reference for remodeling.

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Appendix A: R Code

The code in Listing 1 compiles the simulation for base model. We simulated base model twice with different parameter, but the code are same, just changing the parameter variable at the beginning. In addition, the code in Listing 2 compiles the simulation for extended model.

```
library(ggplot2)
library(tidyr)
library(dplyr)
library(RColorBrewer)

# parameters
b <- 0.0005
d <- 0.0006
beta_i <- 0.1
beta_F <- 0.3
g <- 0.3
sigma <- 0.5

# t: week
t_vals <- seq(1,25, by=1)

# statement variables
S_popn <- rep(NA, times = length(t_vals))
I_popn <- rep(NA, times = length(t_vals))
F_popn <- rep(NA, times = length(t_vals))
R_popn <- rep(NA, times = length(t_vals))

# initial condition
S_popn[1] <- 9.99
I_popn[1] <- 0.01
F_popn[1] <- 0
R_popn[1] <- 0

# model
for (t in 2:length(t_vals)){
  S <- S_popn[t-1] - beta_i * S_popn[t-1]*I_popn[t-1] -
```

```

    beta_F*S_popn[t-1]*F_popn[t-1] + b*S_popn[t-1] - d*S
      _popn[t-1]

I <- I_popn[t-1] + beta_i * S_popn[t-1]*I_popn[t-1] +
  beta_F*S_popn[t-1]*F_popn[t-1] - g*I_popn[t-1] + b*I
    _popn[t-1]

F <- F_popn[t-1] + g*I_popn[t-1] - sigma*F_popn[t-1]

R <- R_popn[t-1] + sigma*F_popn[t-1] + d*S_popn[t-1]

# Set negative values to zero
S_popn[t] <- max(S, 0)
I_popn[t] <- max(I, 0)
F_popn[t] <- max(F, 0)
R_popn[t] <- max(R, 0)
}

df <- data.frame(
  Week = t_vals,
  Susceptible = S_popn,
  Infected = I_popn,
  Funeral = F_popn,
  Removed = R_popn
)

df_long <- pivot_longer(df,
  cols = -Week,
  names_to = "Statement_Variables"
  ,
  values_to = "Population")

ggplot(df_long, aes(x = Week, y = Population, color =
  Statement_Variables)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 1.5) +
  scale_color_brewer(palette = "Dark2") +
  labs(

```

```

    title = "Model_Simulation",
    x = "Week",
    y = "Number_of_Population_(Million)",
    color = "Population"
) +
theme_minimal(base_size = 18) +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"
    ),
  legend.position = "bottom",
  panel.grid.major = element_line(color = "gray"),
)

```

Listing 1: Base Model simulation

```

library(ggplot2)
library(tidyr)
library(dplyr)
library(RColorBrewer)

# parameters
b <- 0.000373
d <- 0.000231
h <- 0.7
beta_i <- 0.1
beta_F <- 0.3
gamma_1 <- 0.7
gamma_2 <- 0.4
sigma_1 <- 0.7
sigma_2 <- 0.5
p1 <- 0.01
p2 <- 0.5

h <- 0.7
p2 <- 0.5
gamma_2 <- 0.4

# t: week
t_vals <- seq(1,30, by=1)

```

```

# statement variables
S_popn <- rep(NA, times = length(t_vals))
E_popn <- rep(NA, times = length(t_vals))
I_popn <- rep(NA, times = length(t_vals))
H_popn <- rep(NA, times = length(t_vals))
F_popn <- rep(NA, times = length(t_vals))
R_popn <- rep(NA, times = length(t_vals))

# initial condition
S_popn[1] <- 9.99
E_popn[1] <- 0
I_popn[1] <- 0.01
H_popn[1] <- 0
F_popn[1] <- 0
R_popn[1] <- 0

# model
for (t in 2:length(t_vals)){
  S <- S_popn[t-1] - beta_i * S_popn[t-1]*I_popn[t-1] -
    beta_F*S_popn[t-1]*F_popn[t-1] + b*S_popn[t-1] - d*S_
    _popn[t-1]

  E <- E_popn[t-1] + beta_F*S_popn[t-1]*F_popn[t-1] +
    beta_i * S_popn[t-1]*I_popn[t-1] - sigma_1*E_popn[t
    -1]

  I <- I_popn[t-1] + sigma_1*E_popn[t-1] - h*I_popn[t-1]
    -
    gamma_1*I_popn[t-1] - p1*I_popn[t-1]+ b*I_popn[t-1]

  H <- H_popn[t-1] + h*I_popn[t-1] - p2*H_popn[t-1] -
    gamma_2*H_popn[t-1]

  F <- F_popn[t-1] + gamma_1*I_popn[t-1] + gamma_2*H_
    popn[t-1] -
    sigma_2*F_popn[t-1]

  R <- R_popn[t-1] + sigma_2*F_popn[t-1] + p1*I_popn[t
    -1] + p2*H_popn[t-1] +
    d*S_popn[t-1]

```

```

# Set negative values to zero
S_popn[t] <- max(S, 0)
E_popn[t] <- max(E, 0)
I_popn[t] <- max(I, 0)
H_popn[t] <- max(H, 0)
F_popn[t] <- max(F, 0)
R_popn[t] <- max(R, 0)
}

df <- data.frame(
  Week = t_vals,
  Susceptible = S_popn,
  Infected = I_popn,
  Funeral = F_popn,
  Removed = R_popn,
  Exposed = E_popn,
  Hospitalized = H_popn
)

df_long <- pivot_longer(df,
  cols = -Week,
  names_to = "Statement_Variables",
  ,
  values_to = "Population")

ggplot(df_long, aes(x = Week, y = Population, color =
  Statement_Variables)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 1.5) +
  scale_color_brewer(palette = "Dark2") +
  labs(
    title = "Model_Simulation",
    x = "Week",
    y = "Number_of_Population_(Million)",
    color = "Population"
  ) +
  theme_minimal(base_size = 18) +

```

```
theme(  
  plot.title = element_text(hjust = 0.5, face = "bold"  
    ),  
  legend.position = "bottom",  
  panel.grid.major = element_line(color = "gray"),  
)
```

Listing 2: Extended Model simulation

Appendix B: Individual Contributions

- Zoey Li: Model Building, Section 2.1 and Section 2.3
- Shiyun Ding: Model Building, Section 2.1 and Section 2.3
- Manru Zhao: Introduction, Section 1, Section 2.2
- Peiyi Wang: Model simulation, Section 2.2, Section 3 and Appendix A
- Jiayi Cai: Future Development, Section 4